Comptia A+ Assignment

Module -1, 2 [Hardware and its components]

Assemble and Disassemble a PC

Assemble and disassemble a desktop PC. Document the process with labeled steps.

Task

❖ **1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.**

➢ List of Main Components Found Inside a Desktop PC

1. Motherboard
   - The motherboard is the main circuit board of the computer. It connects and allows communication between all hardware components.

2. CPU (Central Processing Unit)
   - The CPU is known as the brain of the computer. It processes instructions and performs calculations required for all tasks.

3. CPU Cooler / Heat Sink

- The CPU cooler removes heat generated by the processor. It helps keep the CPU at a safe operating temperature.

4. RAM (Random Access Memory)
   - RAM is temporary memory used while programs are running. It improves the speed and multitasking performance of the computer.

5. Power Supply Unit (PSU)
   - The PSU converts AC electricity into DC power. It supplies power to all internal computer components.

6. Hard Disk Drive (HDD) / Solid State Drive (SSD)
   - HDD or SSD stores the operating system, software, and user files permanently. SSDs are faster than HDDs.

7. Graphics Card (GPU)
   - The graphics card processes images, videos, and games. It improves graphical performance and supports high-resolution displays.

8. Cooling Fan / Cabinet Fan
   - Cabinet fans improve airflow inside the computer case. They help cool internal components and prevent overheating.

9. SATA Data Cable
   - SATA cables connect storage devices such as HDDs and SSDs to the motherboard for data transfer.

10. Power Cables

- Power cables connect the PSU to the motherboard, storage drives, graphics card, and other components.

11. CMOS Battery

- The CMOS battery keeps BIOS settings and the system clock saved even when the computer is turned off.

12. Expansion Slots (PCIe Slots)

- PCIe slots allow additional hardware such as graphics cards, Wi-Fi adapters, or sound cards to be installed.

13. Front Panel Connectors

- These connectors link the cabinet's power button, reset button, USB ports, LEDs, and audio ports to the motherboard.

❖ **2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**

➢ Desktop PC Disassembly Process with explanation

➢ Step 1: Shut Down and Unplug the Computer

- Before starting the disassembly process, I shut down the computer completely and unplugged the power cable. I also disconnected all external devices such as the monitor, keyboard, mouse, and network cable to ensure safety.

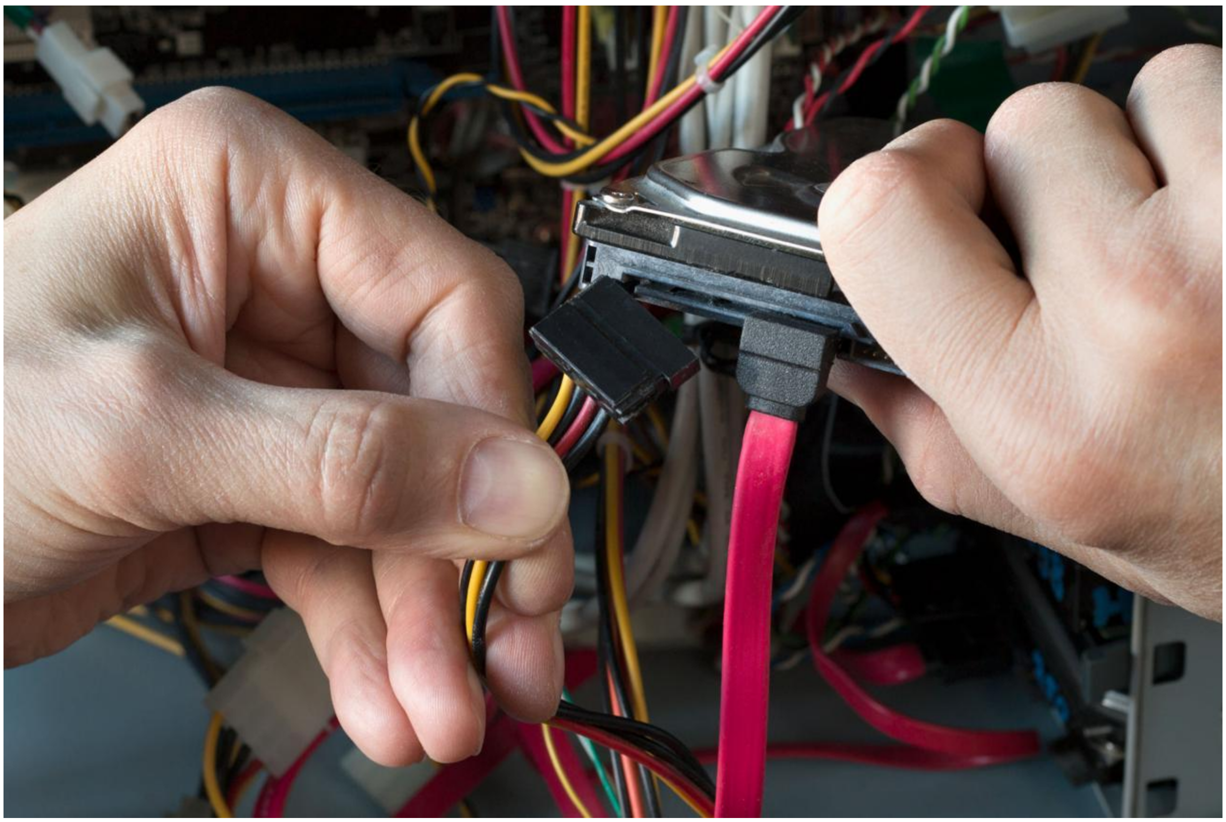
➢ Step 2: Remove the Side Panel

- Using a screwdriver, I removed the screws from the back of the cabinet. Then I carefully slid the side panel backward and lifted it off to access the internal components.

➢ <u>Step 3: Identify the Internal Components</u>

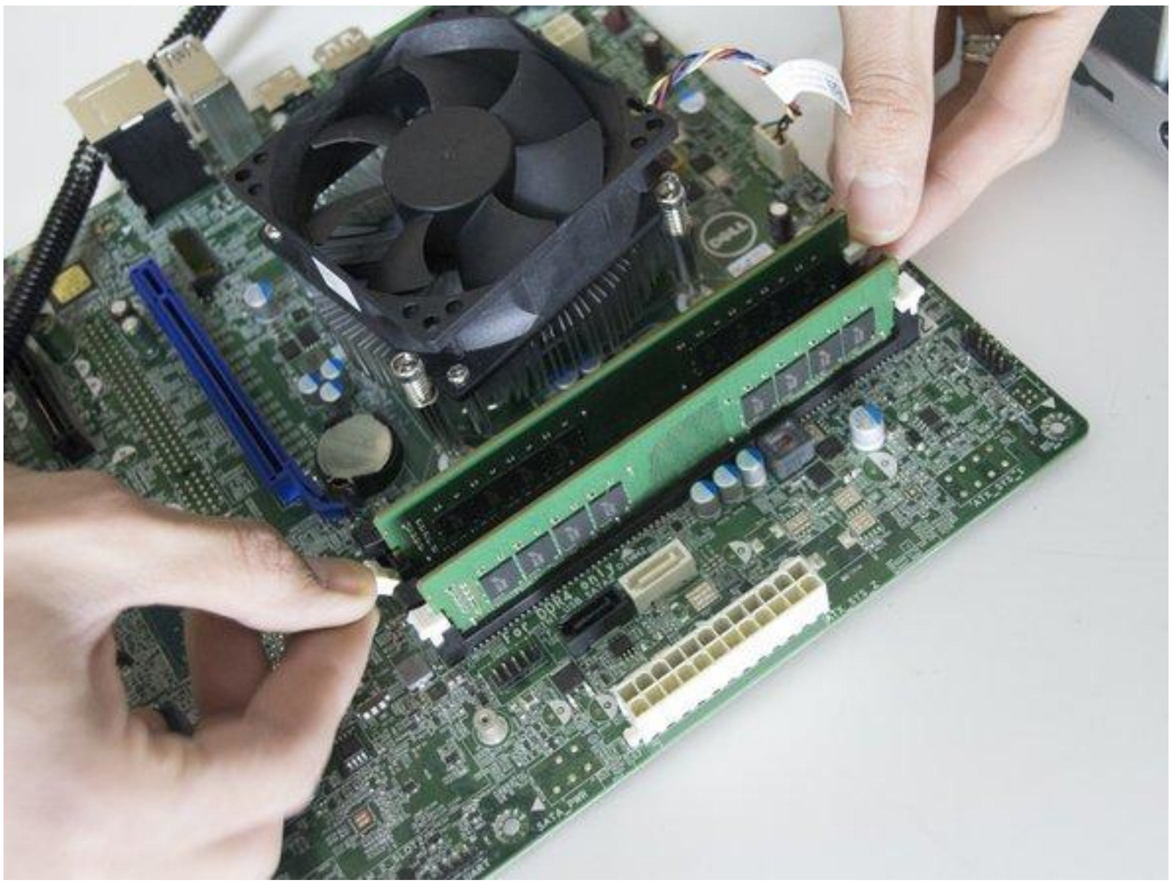
- After opening the cabinet, I identified the major hardware components, including the motherboard, CPU, RAM, power supply (PSU), storage drive (HDD/SSD), cooling fan, and connecting cables.

➢ Step 4: Disconnect Power and Data Cables



- I carefully disconnected all power and SATA data cables connected to the motherboard, storage devices, and power supply. I handled each cable gently to avoid damaging the connectors.

- Step 5: Remove the RAM



- I pressed the retaining clips on both sides of the RAM slot. The RAM module popped up slightly, allowing me to remove it carefully by holding its edges.

➢ <u>Step 6: Remove the HDD/SSD</u>

- After disconnecting the SATA data and power cables, I removed the screws securing the HDD/SSD and carefully lifted the storage device out of the drive bay.

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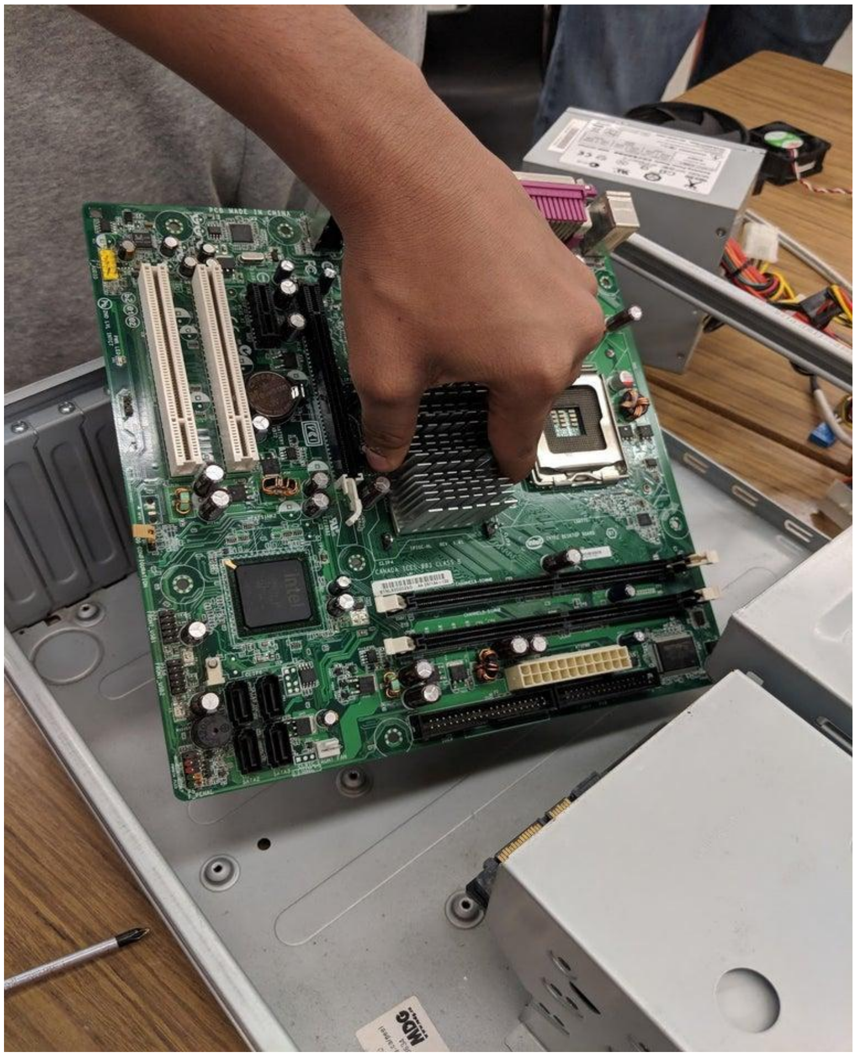
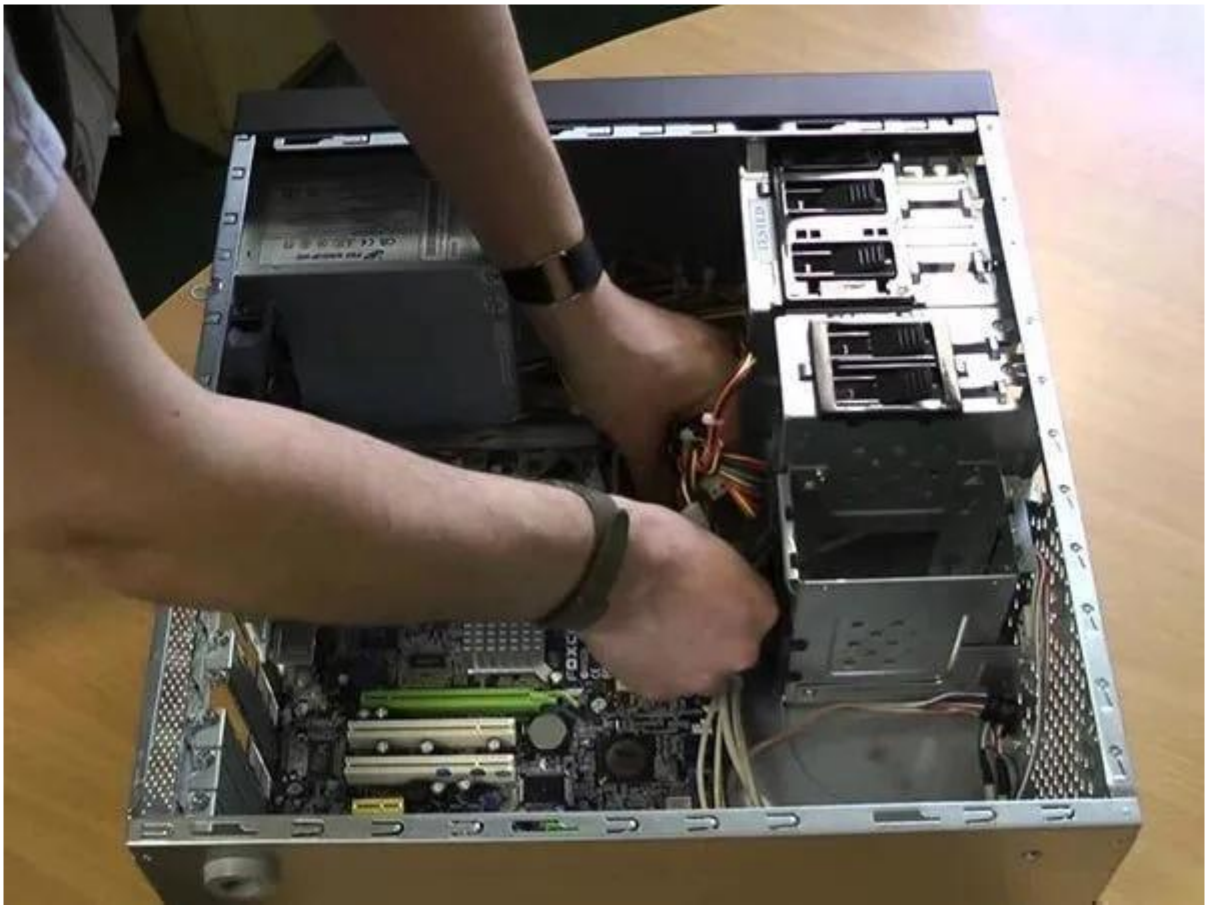
➢ <u>Step 7: Remove the Motherboard</u>



- I disconnected all remaining cables attached to the motherboard and removed the mounting screws. Then I carefully lifted the motherboard out of the cabinet while holding it by the edges.

➢ Step 8: Reassemble the Desktop PC



- After completing the inspection, I reinstalled the motherboard, RAM, HDD/SSD, and all cables in the correct positions. Finally, I attached the side panel, connected the external cables, and powered on the computer to verify that it was working properly.

❖ **3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.**

➢ Reassembling a Desktop PC – Step-by-Step Documentation

➢ Step 1: Install the Motherboard

➢ I carefully placed the motherboard inside the cabinet, aligned it with the standoffs, and secured it using screws.
➢ Why the Order Matters:
➢ The motherboard is the foundation of the computer. It must be installed first because all other components connect to it.

- Step 2: Install the CPU Cooler

- I installed the CPU cooler on the processor and connected the CPU fan cable to the motherboard.
- Why the Order Matters:
- The CPU cooler prevents the processor from overheating, so it must be installed before the computer is powered on.

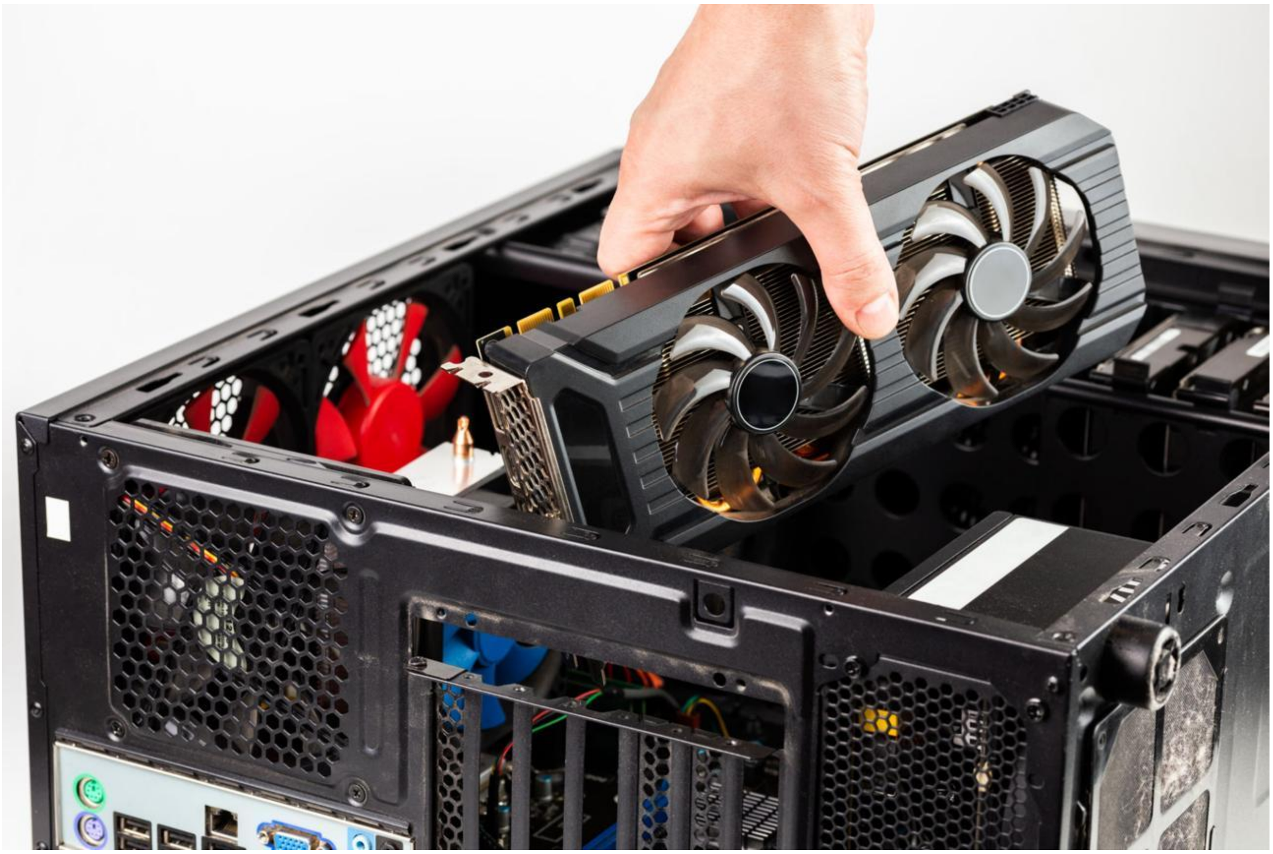
- <u>Step 3: Install the RAM</u>

- I aligned the RAM module with the memory slot and pressed it firmly until both locking clips clicked into place.
- <u>Why the Order Matters:</u>
- RAM is required for the computer to boot and run programs, so it must be installed before starting the system.

- Step 4: Install the HDD/SSD

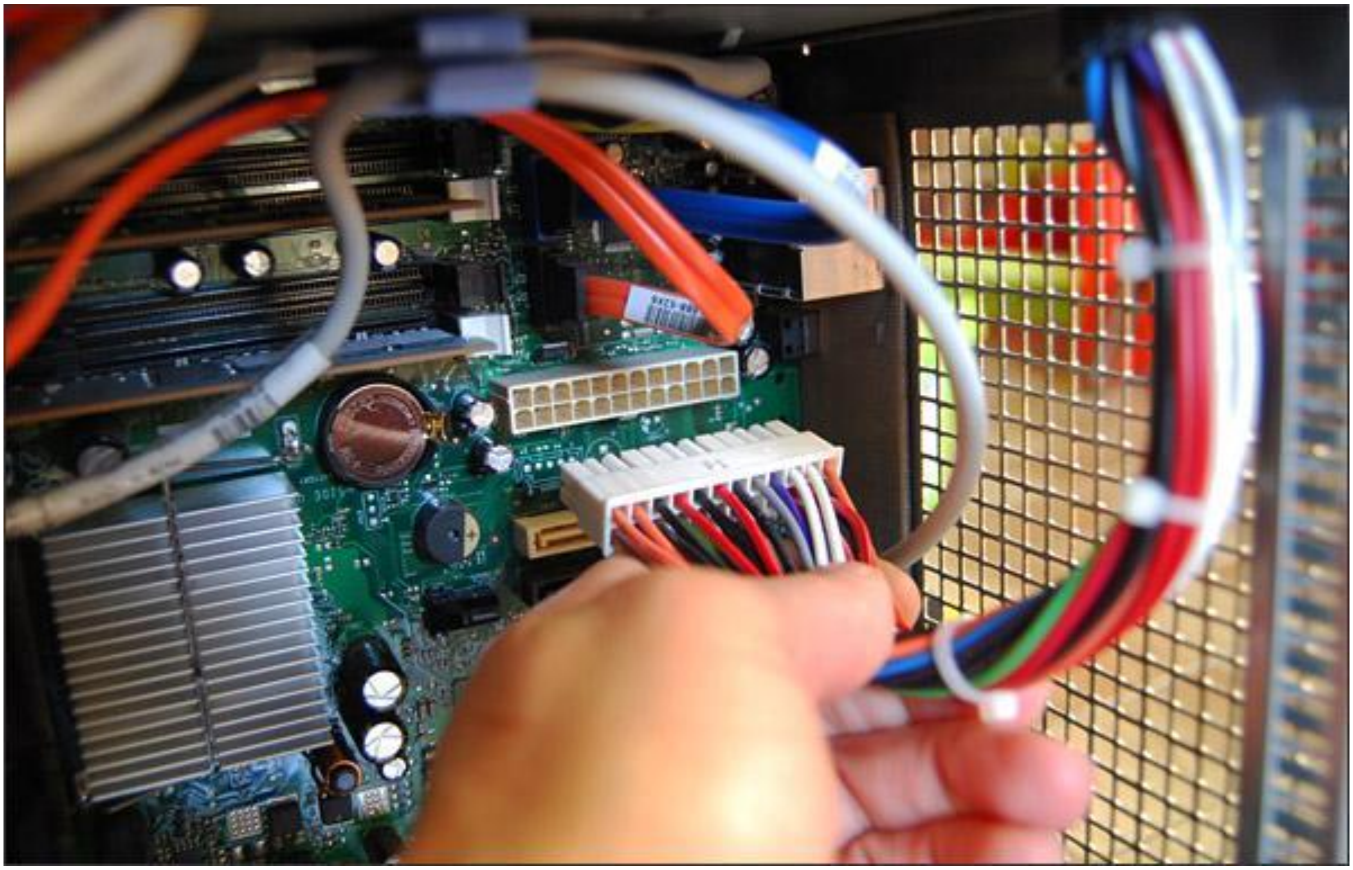
- I mounted the HDD/SSD in the drive bay and connected both the SATA data cable and SATA power cable.
- Why the Order Matters:
- The storage device contains the operating system and files, so it must be connected before booting the computer.

- Step 5: Install the Graphics Card



- I inserted the graphics card into the PCIe slot, secured it with a screw, and connected the PCIe power cable if required.
- Why the Order Matters:
- The graphics card should be installed after the motherboard and RAM to ensure proper alignment and secure installation.

- <u>Step 6: Connect Power Supply Cables</u>



- I connected the 24-pin motherboard power cable, CPU power cable, SATA power cable, and GPU power cable.
- <u>Why the Order Matters:</u>
- Power cables are connected after all hardware is installed to avoid incorrect connections and reduce the risk of damage.

➢ Step 7: Connect Front Panel Connectors

➢ I connected the power switch, reset switch, power LED, HDD LED, front USB, and audio connectors to the motherboard.
➢ Why the Order Matters:
➢ These connections allow the cabinet buttons, USB ports, LEDs, and front audio ports to function correctly.

- <u>Step 8: Organize the Cables</u>

- I arranged all cables neatly to improve airflow and prevent them from touching the cooling fans.
- <u>Why the Order Matters:</u>
- Good cable management improves cooling, reduces dust buildup, and makes future maintenance easier.

- Step 9: Close the Cabinet

- After checking all connections, I reattached the side panel and tightened the screws securely.
- Why the Order Matters:
- Closing the cabinet protects internal components from dust, accidental contact, and physical damage.

- Step 10: Test the Computer

- I connected the monitor, keyboard, mouse, and power cable. I switched on the computer and confirmed that it booted successfully.
- Why the Order Matters:
- Testing at the end verifies that all components have been installed correctly and the computer is functioning properly.

❖ 4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

➢ Safety Precautions Checklist

Safety Precaution	Why It Is Important
Turn off the computer before opening the cabinet.	Prevents electric shock and protects hardware from damage.
Unplug the power cable from the wall socket.	Ensures that no electrical current is flowing through the system.
Disconnect all external devices.	Prevents accidental damage to connected devices and makes handling easier.
Wear an anti-static wrist strap or ground yourself.	Protects electronic components from electrostatic discharge (ESD).
Handle components by their edges only.	Avoids damage to chips, circuits, and gold contacts.
Keep the work area clean and dry.	Reduces the risk of dust, moisture, and accidental damage.
Use the correct screwdriver size.	Prevents stripped screws and damage to computer components.
Do not force components into their slots.	Components fit only in the correct orientation; forcing them may cause permanent damage.
Keep screws and small parts organized.	Prevents losing important parts and makes reassembly easier.
Check all cable connections before powering on.	Ensures the system starts safely and all components function correctly.

- Tools needed before starting the PC

Tool	Why It Is Important
Phillips Screwdriver	Used to remove and tighten cabinet, motherboard, and other component screws safely.
Flat-head Screwdriver	Useful for certain screws and gently releasing clips when required.
Anti-Static Wrist Strap	Prevents electrostatic discharge (ESD), which can damage sensitive electronic components.
Thermal Paste	Applied between the CPU and CPU cooler to improve heat transfer and prevent overheating.
Cable Ties	Help organize cables inside the cabinet for better airflow and easier maintenance.
Small Container for Screws	Keeps screws and small parts safe, preventing them from getting lost during the process.
Soft Cleaning Brush	Removes dust from components without damaging delicate parts.
Microfiber Cloth	Used to clean the cabinet and components without leaving scratches or lint.